

VARADHA VR

Robotics Systems Engineer

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PROFESSIONAL SUMMARY

Robotics Systems Engineer with **2+ years of industry experience** developing and deploying robotics and IoT solutions. Experienced in industrial cobots, hardware-software integration, and system architecture.

TECHNICAL SKILLS

- **Robotics** : Industrial Robotic Arms (Doosan, Epson, SCARA) · ROS2 · DART · Gazebo
- **Embedded/IoT** : ESP32 · Raspberry Pi · STM32 · I2C · SPI · UART · MQTT · WebSocket · REST APIs · Firebase
- **Programming** : Python · C++
- **Computer Vision**: OpenCV · MediaPipe
- **Systems Engineering**: Requirements Management · System Architecture · Hardware-Software Integration · Technical Documentation

WORK EXPERIENCE

Robotics Systems Engineer

Inker Robotics, Thrissur, Kerala | July 2024 – September 2026

- Designed and developed **end-to-end robotics and IoT solutions**, covering system architecture, hardware-software integration, testing, debugging, and deployment.
- Programmed, commissioned, and deployed **industrial robotic arms**, developing automation solutions for real-world applications.
- Analyzed **client requirements** and translated them into system architectures, technical solutions, development workflows, and deployment plans.
- Built and validated **Proof-of-Concept systems** across IoT, and robotics domains to evaluate and demonstrate real-world applications.
- Led the design and development of **Robopark's smart ticketing platform**, covering user journeys, booking architecture, biometric authentication, system workflows, backend requirements, API integration, and notification logic.
- Owned the **IoT domain within the Academic Department**, developing technical curricula and conducting internships, industrial visits, workshops, FDPs, and hands-on technical sessions for engineering students and faculty.

PROJECTS

Robobooth – Automated Robotic Videography System

- Architected system with **Doosan 6-axis collaborative robotic arm**, camera control, and backend orchestration.
- Developed a **FastAPI backend** coordinating Doosan arm motion through the DART platform and Android camera recording through ADB commands.
- Designed hardware triggering using **Arduino-based relay control**, synchronizing robotic motion and camera operation.
- Automated **FFmpeg-based video processing** and deployed to production operation with automated session handling and rapid video turnaround.

AIoT Smart School Campus Monitoring System

- Designed a modular IoT platform for real-time monitoring of **attendance, occupancy, environmental conditions, energy consumption, parking, and emergency-response systems**.
- Developed a web-based **3D digital-twin interface using React and Three.js** for visualizing campus systems and live sensor data.
- Integrated ESP32-based sensing nodes with the web application using WebSocket communication for real-time system visualization and control.

EDUCATION

B.Tech – Electronics & Communication Engineering

Cochin University of Science and Technology (CUSAT), Kochi | Graduation: **2024** | **CGPA: 8.8/10**